

Lab07

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Lab07

```
library(DBI)
library(RSQLite)

fname <- "database.sqlite3"
con <- dbConnect(RSQLite::SQLite(), fname)
```

```
SELECT COUNT(DISTINCT i.id) AS total_professores
FROM instructors i
JOIN teachings t ON i.id = t.instructor_id
JOIN sections sec ON t.section_uuid = sec.uuid
JOIN subject_memberships sm ON sec.course_offering_uuid = sm.course_offering_uuid
JOIN subjects s ON sm.subject_code = s.code
WHERE s.abbreviation = 'STAT';
```

Table 1: 1 records

total_professores
379

```
SELECT DISTINCT i.name AS professor
FROM instructors i
JOIN teachings t ON i.id = t.instructor_id
JOIN sections sec ON t.section_uuid = sec.uuid
JOIN subject_memberships sm ON sec.course_offering_uuid = sm.course_offering_uuid
JOIN subjects s ON sm.subject_code = s.code
WHERE s.abbreviation = 'STAT'
ORDER BY i.name;
```

Table 2: Displaying records 1 - 10

professor
ABIGAIL BENZINE
ADIN-CRISTIAN ANDREI
AKICHIKA OZEKI
ALAN HUANG
ALBERTO DEL PIA
ALBRECHT KLEMM
ALEXANDER COVINGTON
ALEXANDER FISH
ALEXANDER KISELEV
ALYSSA DIGILIO

```

SELECT
  i.name AS professor,
  ROUND(
    SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 + gd.bc_count * 2.5 + gd.c_count * 2.0 + gd.d_count * 1.5 + gd.f_count * 1.0) /
    SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.f_count)
  ) AS gpa_medio
FROM instructors i
JOIN teachings t ON i.id = t.instructor_id
JOIN sections sec ON t.section_uuid = sec.uuid
JOIN grade_distributions gd ON sec.course_offering_uuid = gd.course_offering_uuid AND sec.number = gd.number
JOIN subject_memberships sm ON sec.course_offering_uuid = sm.course_offering_uuid
JOIN subjects s ON sm.subject_code = s.code
WHERE s.abbreviation = 'STAT'
GROUP BY i.id, i.name
HAVING SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.f_count) > 0
ORDER BY gpa_medio ASC
LIMIT 1;

```

Table 3: 1 records

professor	gpa_medio
JAMES D KUELBS	2.638

```

SELECT
  i.name AS professor,
  ROUND(

```

```

SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 + gd.bc_count * 2.5 + gd.c_count * 2.0 + gd.d_count + gd.f_count)
SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.f_count)
) AS gpa_medio
FROM instructors i
JOIN teachings t ON i.id = t.instructor_id
JOIN sections sec ON t.section_uuid = sec.uuid
JOIN grade_distributions gd ON sec.course_offering_uuid = gd.course_offering_uuid AND sec.num_students > 0
JOIN subject_memberships sm ON sec.course_offering_uuid = sm.course_offering_uuid
JOIN subjects s ON sm.subject_code = s.code
WHERE s.abbreviation = 'STAT'
GROUP BY i.id, i.name
HAVING SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.f_count) > 0
ORDER BY gpa_medio DESC
LIMIT 1;

```

Table 4: 1 records

professor	gpa_medio
SRIKANTHMADHAVAN ARAVAMUTHAN	4

```

SELECT
co.name AS disciplina,
ROUND(
SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 + gd.bc_count * 2.5 + gd.c_count * 2.0 + gd.d_count + gd.f_count)
SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.f_count)
) AS gpa_medio
FROM course_offerings co
JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects s ON sm.subject_code = s.code
WHERE s.abbreviation = 'STAT'
GROUP BY co.name
HAVING SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.f_count) > 0
ORDER BY gpa_medio ASC
LIMIT 1;

```

Table 5: 1 records

disciplina	gpa_medio
Intro-Theory of Probability	2.886

```

SELECT
  co.name AS disciplina,
  ROUND(
    SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 + gd.bc_count * 2.5 + gd.c_count * 2.0 + gd.d_count * 1.5 + gd.e_count * 1.0 + gd.f_count * 0.5) /
    SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.e_count + gd.f_count)
  ) AS gpa_medio
FROM course_offerings co
JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects s ON sm.subject_code = s.code
WHERE s.abbreviation = 'STAT'
GROUP BY co.name
HAVING SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.e_count + gd.f_count) > 0
ORDER BY gpa_medio DESC
LIMIT 1;

```

Table 6: 1 records

disciplina	gpa_medio
Sample Survey Theory&Method	4

```

Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()

```

```
[1] "2026-09-24 00:51:31 -03"
```